

Masters Program in Advanced Cyber Security

Module -1 Cybersecurity for beginners

Begginers Level

- Introduction to Network, Network Topologies
 - Understanding OSI Model
 - TCP 3 way Handshake, TCP vs UDP
 - IP address, mac address and subnetting explained
 - Elements of Information Security
 - Authentication and Multi Factor Authentication
 - Data and Security Controls
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- Motives, Top Infosec Threats and Vectors
 - What is Hacking, Phases of hacking
 - IOCs, Information Security Laws and Strategies
 - What is Cryptography, Symmetric vs Asymmetric
 - Setting Up Virtual Box, Kali Linux OS
 - Setting Up Windows and Ubuntu
 - Basics of Linux - How to Use, Commands
 - Mapping and Scanning the Network with Nmap
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Intermediate level

- Motives, Top Infosec Threats and Vectors
 - What is Hacking, Phases of hacking
 - IOCs, Information Security Laws and Strategies
 - What is Cryptography, Symmetric vs Asymmetric
 - Setting Up Virtual Box, Kali Linux OS
 - Setting Up Windows and Ubuntu
 - Basics of Linux - How to Use, Commands
 - Mapping and Scanning the Network with Nmap
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Advanced level

- Vulnerability Scanning With Nessus
 - Linux Hacking & Password Cracking with Metasploit
 - Dos and DDos Attack Explained with Practical
 - MITM Attack Explained with Practical
 - Basics of web application
 - OWASP TOP 10, Checklist, Hacktricks
 - Basics of Burpsuite, Installing and Configuring
 - Installing OWASP JUICE Shop, DVWA Using Docker
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Expert level

- How to Bruteforce a Login Page Using Burp Suite
 - Exploiting File Upload Vulnerability
 - Exploiting SQL Injection Vulnerability
 - What is Bug Bounty, How to Find Bugs
 - Logs, SOC, SIEM Basics
 - What is Mitre Attack Framework Explained
 - Wireshark, Explained with Practical
 - Incident Response Explained
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- Project Explanation and hints

Module -2

Advanced Cybersecurity

Begginer level

- Network Security - Threats & Access Controls
 - Network Security Architecture - VPN and Wireless Security
 - Network Security Architecture - Firewall
 - Network Security Architecture - IDS & IPS
 - Network Attacks - Most Common Types
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Intermediate Level

- Network Attacks - Reconnaissance
 - Network Attacks - Access Attacks
 - Network Attacks - Service Attacks
 - Network Attacks -Malware
 - Network Attacks - Advanced Persistent Threat
 - Network Attacks - Countermeasures
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Advanced Module

- Network Security Monitoring - Active Monitoring Part one
 - Network Security Monitoring - Active Monitoring Part Two
 - Network Security Monitoring - Log Maintenance and Monitoring
 - Web Application Security - OWASP Lab Setup
 - Web Application Security - OWASP Top 10 - Sql injection
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Expert Module

- Web Application Security - OWASP Top 10 - Broken Authentication
 - Web Application Security - OWASP Top 10 - Sensitive Data Exposure
 - Web Application Security - OWASP Top 10 - Broken Access Control
 - Web Application Security - OWASP Top 10 - Security Misconfiguration & XSS
 - Web Application Security - OWASP Top 10 - Insecure deserialization and More
 - Web Application Security - General Threat and Attacks
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Module - 3

Advanced Ethical Hacking

Begginer Level

- BurpSuite Professional Installation
 - BurpSuite Walkthrough
 - Broken Access Control - 1
 - Broken Access Control - 2
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Intermediate module

- Broken Access Control - 3
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- Broken Access Control - 4
 - Cryptographic Failures
 - Cross-Site Scripting Injection (XSS)
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Advanced Module

- SQL Injection - 1
 - SQL Injection - 2
 - SQL Injection - 3
 - OS Command Injection
 - File Upload Vulnerability
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Expert Module

- Cross-Site Request Forgery (CSRF)
 - Business Logic Flaws
 - Low Severity Vulnerabilities
 - Identification and Authentication Failure - 1
 - Identification and Authentication Failure - 2
 - Module - 4
 - Mobile Hacking
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Beginners Level

- DEMO Video
 - How to install kali linux(the easiest way)
 - Linux Basics
 - Installing BlueStacks as our android emulator
 - Introduction to Mobile Hacking
 - What is Android and it's architecture
 - What is apple iOS and it's architecture
 - Introduction to Mobile Security Landscape
 - Types of Mobile Attacks
 - Understanding MITM attack
 - MITM attack Practical
 - Understanding Physical Security Attacks
 - Understanding OWASP top 10 Mobile Risks
 - Vulnerabilities in Mobile Apps
 - Future Trends and improvements, IoT Devices
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Intermediate level

- What is Termux and it's features
 - Configuring and install Termux
 - Personalizing Termux - Hacker style
 - Install 540+ tools in termux using darkfly
 - Using Seeker To Hack Geolocation Part 1
 - Using Seeker To Hack Geolocation Part 2
 - Introduction to Social Engineering
 - What are Human Based Social Engineering Techniques
 - What are Computer Based Social Engineering Techniques
 - What are Mobile Based Social Engineering Techniques
 - Using Zphisher for Phishing - Automated Phishing tool
 - Social Engineering Toolkit Tutorial
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Advanced Module

- Introduction to Information Security - CIA triad
 - What is BYOD Policy - Explained
 - What is Mobile Device Management
 - Metasploit Framework - Explained
 - How To generate a payload for mobile hacking
 - Taking control of device using msf
 - Hacking android using adb port using phonesploit
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Expert Level

- Introduction to Reverse Engineering
 - How to analyze apps using online softwares
 - Using apk editor to edit apps
 - Best Practices - Device and Network Configurations
 - What is Call forwarding, Secret Codes
 - Legal Guidelines and ethical use
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Assignment 1

Assignment 2

Case Studies - 10 Real time Projects